

Pegah Mohammadipour

Education	Ph.D. in Mathematics	August 2022 - Present
	The Pennsylvania State University, PA, USA	
	<i>Advisor: Prof. Xiantao Li</i>	
	<i>Focus: Quantum Algorithms for Open Quantum Systems</i>	
	B.Sc. in Mathematics and Applications	September 2017 - September 2021
	University of Tehran, Tehran, Iran	
	<i>Undergraduate Thesis: The Geometric Concept of Parallel Transport along Curves with some Physical Interpretations</i>	
	Minor in Computer Engineering	February 2019 - August 2021
	University of Tehran, Tehran, Iran	
	Diploma in Mathematics and Physics	September 2010 - July 2017
	Aboureihan Educational Complex, Tehran, Iran	
Publications	• Pegah Mohammadipour, X. Li. <i>MLMC-qDRIFT: Multilevel Variance Reduction for Randomized Quantum Hamiltonian Simulation</i>, <i>arXiv</i>, doi: 0.48550/arXiv.2604.26865. • Pegah Mohammadipour, X. Li. <i>Reducing Circuit Depth in Lindblad Simulation via Step-Size Extrapolation</i>, <i>Physical Review A</i>, 112(6):062206, 2025, doi: 10.1103/dzpc-m1p4. • Pegah Mohammadipour, X. Li. <i>Direct Analysis of Zero-Noise Extrapolation: Polynomial Methods, Error Bounds, and Simultaneous Physical-Algorithmic Error Mitigation</i>, <i>Quantum</i>, 9, 1909 (2025)., doi: 10.22331/q-2025-11-14-1909. • H. Mahdavi, P. Mahdavinia, A. Farhadi, Pegah Mohammadipour, S. Malek, P. Mohammadipour, M. Daliri, A. Hashemi, A. Khasahmadi, V. Honavar. <i>CombiGraph-Vis: A Multimodal Olympiad Benchmark for Discrete Mathematical Reasoning</i>, <i>NeurIPS 2025 Workshop MATH-AI</i> arXiv:2510.27094v1., • H. Mahdavi, P. Mahdavinia, S. Malek, Pegah Mohammadipour, S. Malek, P. Mohammadipour, A. Hashemi, M. Daliri, A. Farhadi, A. Khasahmadi, N. Mireshghallah, V. Honavar. <i>RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows</i>, <i>NeurIPS 2025 Workshop MATH-AI</i>, arXiv:2510.09021v1. • H. Mahdavi, A. Hashemi, M. Daliri, Pegah Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, V. Honavar. <i>Brains vs. Bytes: Evaluating LLM Proficiency in Olympiad Mathematics</i>, <i>COLM Conference, 2025</i>, arXiv:2504.01995v2.	
Experience	Research Intern	June 2026 2021 - August 2026
	Fujitsu Research America, Santa Clara, USA	
	<i>Developing quantum algorithms for finance</i>	
	Research Assistant	September 2021 - May 2022
	Institute for Research in Fundamental Sciences (IPM), Tehran, Iran	

Functional Inequalities for Quantum Correlations in Networks
Advisor: Dr. Salman Beigi

Research Assistant July 2019 - September 2020
School of Electrical and Computer Engineering, University of Tehran,
Tehran, Iran
Deep learning-based image compression
Advisor: Dr. Farhad Pakdaman

Talks **Contributed Talk, APS Global Summit** March 2026
Reducing Circuit Depth in Lindblad Simulation via Step-Size Extrapolation

Lightning Poster Round, IPAM Workshop on New Frontiers in Quantum Algorithms for Open Quantum Systems January 2026
Polynomial Extrapolation for Lindblad Simulation

Contributed Talk, SIAM New York-New Jersey-Pennsylvania Section November 2025
Polynomial Extrapolation for Error Mitigation in Quantum Simulation

Professional Service • **Reviewer**, 26th Asian Quantum Information Science Conference (AQIS) 2026

Teaching Experience **Graduate Teaching Associate** August 2022 - May 2025
Department of Mathematics, The Pennsylvania State University, PA, USA

- MATH 231 - Multivariable Calculus

Teaching Assistant September 2019 - January 2022
School of Mathematics, Statistics and Computer Science, University of Tehran,
Tehran, Iran

- *Fall 2021*: Differential Equations
- *Spring 2020*: Calculus 2
- *Fall 2020*: Data Structures
- *Spring 2020*: Calculus 2
- *Fall 2019*: Introduction to Computer Programming

English Teacher March 2018 - October 2018
Safir Language Academy, Tehran, Iran

Professional Activities **APS Global Physics Summit 2026** March 15-20, 2026
Colorado Convention Center, Denver, CO, and online

New Frontiers in Quantum Algorithms for Open Quantum Systems Workshop
Institute for Pure and Applied Mathematics (IPAM),
Los Angeles, CA, USA January 12 - 16, 2026

QSIM Conference IBM, New York City, NY, USA	August 2025
QSIM Conference and Summer School University of Rhode Island, Kingston, Rhode Island, USA	August 2024
Graduate Summer School in Stochastic Quantization SLMath, Berkeley, California, USA	July 2024
Graduate Summer School in Concentration Inequalities and Localization Techniques in High Dimensional Probability and Geometry SLMath, Berkeley, California, USA	July 2023
Undergraduate Summer School on Experimental QIP (USEQIP) Remote, University of Waterloo, Canada	May 2021 - July 2021
Qiskit Global Summer School 2021: Quantum Machine Learning Remote, IBM Quantum	July 2021
Qiskit Global Summer School 2020: Introduction to Quantum Computing and Quantum Hardware Remote, IBM Quantum	July 2020
Game Theory Summer School Institute for Research in Fundamental Sciences (IPM), Tehran, Iran	August 2019
Deep Learning Summer School (UTDLSS) School of Electrical and Computer Engineering, University of Tehran	August 2018

Skills

- **Languages:** Persian(Native), English(Fluent), Spanish(Intermediate)
- **Programming Languages:** C, C++, Python, R, MATLAB, Octave, HTML/CSS
- **Software & Tools:** Qiskit, L^AT_EX, Jupyter Notebooks
- **Operating Systems:** Ubuntu, Windows

Leadership & Volunteering

- **Organizer, Quantum Computation Reading Group**
Penn State University, State College, PA, USA
Fall 2025
- **Poster Judge, ECoS Undergraduate Research Exhibition**
Penn State University, State College, PA, USA
Fall 2024
- **Editor-in-Chief, *Jonge Riazi* Student Mathematics Magazine**
School of Mathematics, Statistics, and Computer Science, University of Tehran, Tehran, Iran
January 2020 – September 2021
 - Revitalized the mathematics magazine after a 24-year hiatus.
 - Recruited and mentored a team of students to develop, edit, and finalize article content.
- **Elected Committee Member, University of Tehran Mathematics Student Association**
University of Tehran, Tehran, Iran
September 2018 – September 2020

- Organized *Math Suppers*, a weekly series of talks on mathematical topics and their applications.
- Created and coordinated student-led mathematics study groups and workshops.

Awards & Honors

- **Honorable Mention**, Mary Lister McCammon Award for Distinguished Undergraduate Teaching
- **Graduate Teaching Associate Status**, awarded in recognition of commitment to undergraduate teaching
- **2nd Place**, Penn State ICDS Symposium Poster Competition 2025
- **Top 1% Rank** in the 2017 Iranian National University Entrance Examination (*Concour*) in Mathematics and Physics